Set Theory — Complete Foundations and CS Applications

A 1.5–2 hour lecture covering notation, membership, subsets, operations, cardinality, power sets, Cartesian products, identities and Python.

• Recommended teaching duration: approximately 1.5–2 hours including board work, worked examples, Python demonstrations and student discussion.

• Teaching approach: concept → notation → worked example → CS interpretation → Python verification → student practice.

Lecture Roadmap

• Concept introduction and terminology

• Detailed definitions and notation

• Worked mathematical examples

• CS case studies with complete solutions

• Python implementation and verification

• Extended question bank

• 2–3 unsolved student practice questions at the end

1. What Is a Set?

A set is a well-defined collection of distinct objects. Objects are called elements. Order does not matter and repetition does not create a new element.

Examples: A={1,2,3}; programming_languages={Python,Java,C++}.

2. Membership and Notation

$x \in A$ means x belongs to A; $x \notin A$ means it does not. Sets may be described by listing elements or by a rule, called set-builder notation.

3. Empty and Universal Sets

The empty set ∅ has no elements. A universal set U contains all objects currently under discussion. Complement A' or A^c contains elements of U not in A.

4. Subsets and Proper Subsets

$A \subseteq B$ means every element of A is in B. A proper subset $A \subset B$ is a subset that is not equal to B. The empty set is a subset of every set.

5. Set Operations

Union $A \cup B$, intersection $A \cap B$, difference $A - B$, symmetric difference, and complement. Draw Venn diagrams to interpret each operation.

Useful identity: $|A \cup B| = |A| + |B| - |A \cap B|$.

6. Cartesian Product

$A\times B$ contains ordered pairs (a,b). In general $A\times B$ is different from $B\times A$. If $|A|=m$ and $|B|=n$, then $|A\times B|=mn$.

7. Power Set

$P(A)$ is the set of all subsets of A. If $|A|=n$, then $|P(A)|=2^n$. For $A=\{a,b\}$, $P(A)=\{\emptyset,\{a\},\{b\},\{a,b\}\}$.

8. Cardinality and Counting

Cardinality is the number of elements in a finite set. Inclusion-exclusion prevents double counting when sets overlap.

9. Set Identities

Commutative, associative, distributive, identity, domination, idempotent, complement, and De Morgan laws can be verified using Venn diagrams or membership reasoning.

De Morgan: $(A\cup B)^c=A^c\cap B^c$ and $(A\cap B)^c=A^c\cup B^c$.

10. Python Sets

Python sets implement mathematical set operations. A set removes duplicates automatically. This is useful for unique identifiers, tags, permissions, and filtering.

11. CS Case Study — Course Registration

P={students in Programming}, D={students in Database}, S={students in Discrete Structures}. Find $P\cap D$, $P\cup D$, $P-D$ and $P\cap D\cap S$.

Solution example: if P={Ali,Sara,Hamza}, D={Sara,Hamza,Ayesha}, S={Sara,Hamza}, then $P\cap D$={Sara,Hamza}; $P-D$={Ali}; $P\cap D\cap S$={Sara,Hamza}; $P\cup D$={Ali,Sara,Hamza,Ayesha}.

12. CS Case Study — Cybersecurity Roles

Admin, Developer and Auditor are sets of users with each role. Find users who are Developer AND Auditor, all privileged users, and Developers who are not Auditors.

Solution: use intersection, union and difference respectively. With Admin={Ali,Sara}, Developer={Sara,Hamza}, Auditor={Sara,Ayesha}, results are {Sara}, {Ali,Sara,Hamza,Ayesha}, and {Hamza}.

13. CS Case Study — Database Tags

A document retrieval system stores tags for documents. Set operations are used to answer “documents containing both Python and AI,” “Python or database,” and “Python but not database.”

Solution: intersection answers AND-tag queries; union answers OR-tag queries; difference removes unwanted categories.

Python Laboratory

Python Activity 1

```
A={'Python','AI','Database'}
B={'Python','Web','Database'}

print("Union:", A | B)
print("Intersection:", A & B)
print("A-B:", A - B)
print("Symmetric difference:", A ^ B)
```

Ask students to predict the output before executing the code, then change at least one input and explain the new result.

Python Activity 2

```
# Remove duplicate IDs
ids=[101,102,101,103,102,104]
unique_ids=set(ids)
print(unique_ids)
```

Ask students to predict the output before executing the code, then change at least one input and explain the new result.

Python Activity 3

```
# Power set for a small set
from itertools import combinations
A={1,2,3}
items=list(A)
power=set()
for r in range(len(items)+1):
    for c in combinations(items,r):
        power.add(frozenset(c))
print(power)
```

Ask students to predict the output before executing the code, then change at least one input and explain the new result.

Extended Question Bank — Instructor-Led Practice

The following questions are designed to provide enough board/classroom practice for a 1.5–2 hour session. They cover the same topic areas as the relevant VU MTH202 sequence, but are written as fresh exercises rather than reproducing the handout wording.

1. Write a set containing five programming languages.
2. Determine membership for selected elements.
3. Find union, intersection and difference of two given sets.
4. Determine whether one set is a subset of another.
5. Find the power set of a two-element set.
6. Find the Cartesian product of two small sets.
7. Verify De Morgan's law for two finite sets.
8. Use inclusion-exclusion to count students enrolled in two courses.
9. Model database tags using sets.
10. Model cybersecurity roles using sets.

11. Write Python code for union, intersection and difference.

12. Write Python code that removes duplicate student IDs.

13. Explain why order is irrelevant in a mathematical set but important in a list.

End-of-Lecture Student Practice — No Solutions

1. A university has four course-enrollment sets. Use inclusion-exclusion to determine the number of students in at least one course.

2. Find the power set of a three-element set and explain its cardinality.

3. Write Python for a real dataset of your choice.

Quick Recap

Sets model collections and provide operations for data organization.